February 17, 2026

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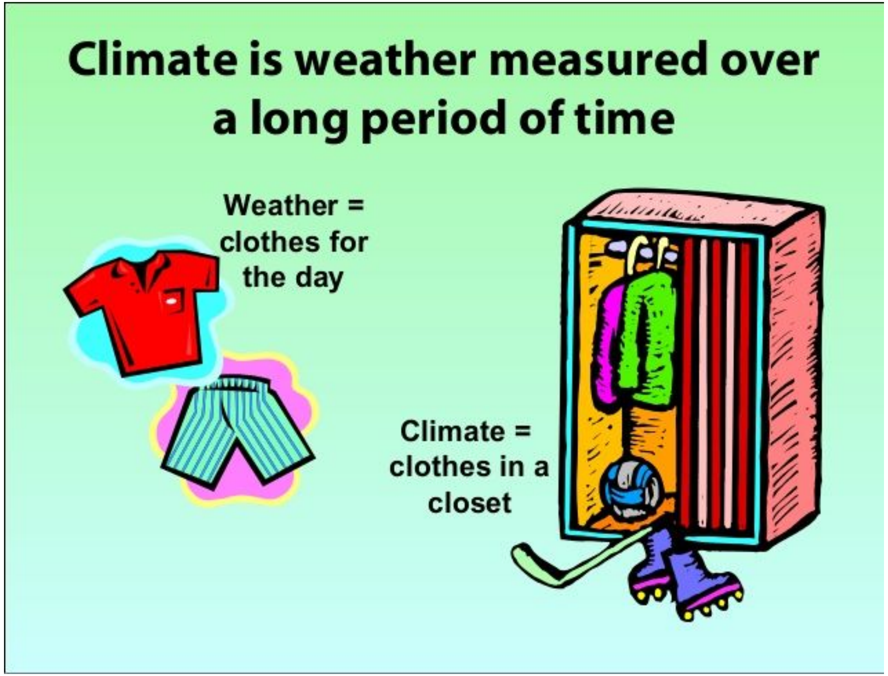
SURPRISE VOCABULARY WORDS

November 8 2021

weather - the condition of the atmosphere at a particular place and time. includes wind, temperature, humidity, atmospheric pressure, cloudiness, and precipitation

climate - the average weather conditions in a particular location or region at a particular time of the year



L3 Vocabulary

May 5, 2025

Tropic of Cancer - the most northerly circle of latitude on Earth at which the Sun can be directly overhead

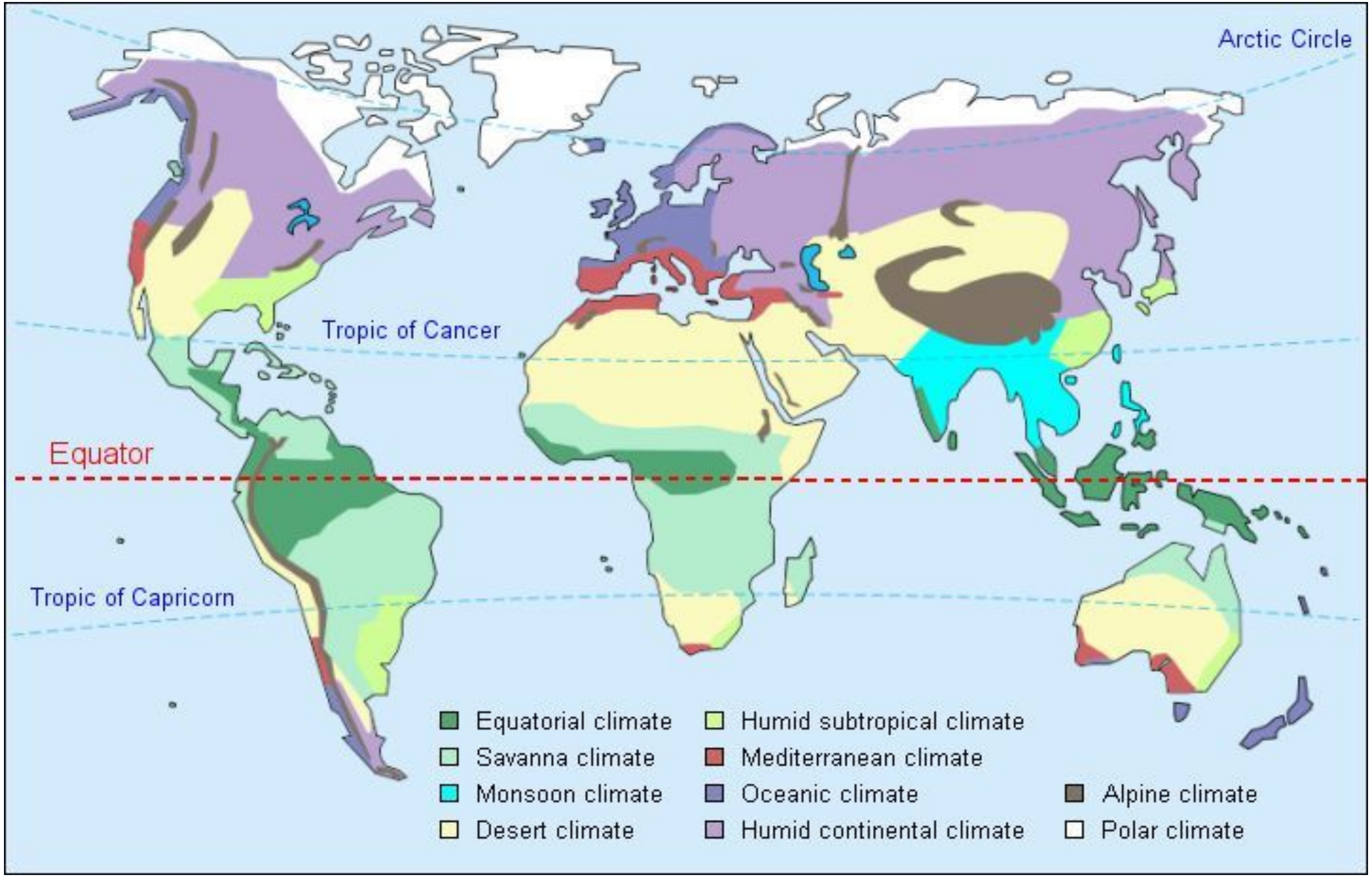
Tropic of Capricorn - the most southerly circle of latitude on Earth at which the Sun can be directly overhead

tropics - the region between the tropics of Cancer and Capricorn

latitude - distance from the equator (North or South)

longitude - distance East or West from the Prime Meridian

World Climates



Factors that determine climate (*the three "tudes"*):

Latitude #1 factor- angle of the sun's rays

Longitude Eastern or Western location on a landmass

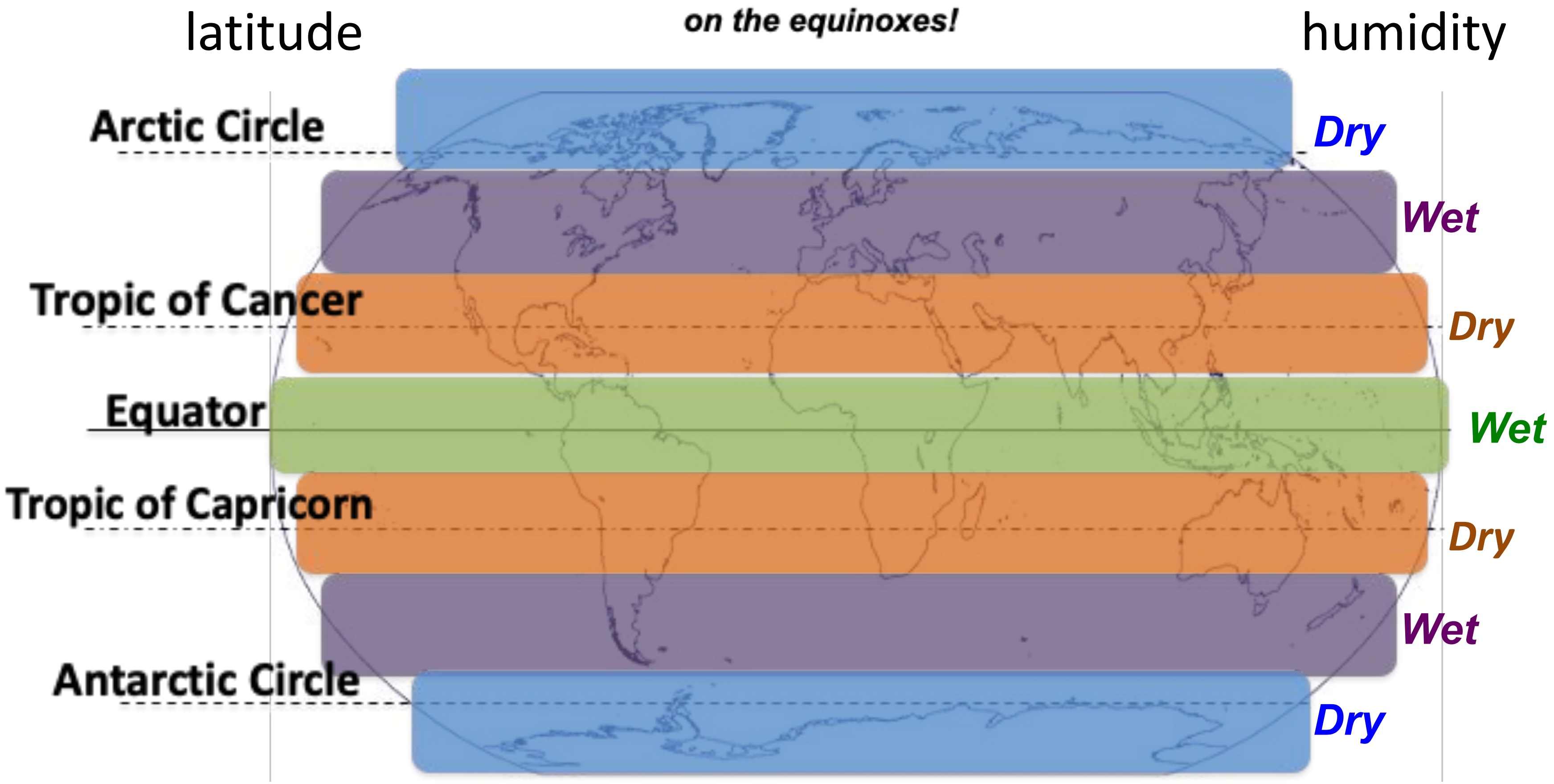
Are you in New York or San Francisco?

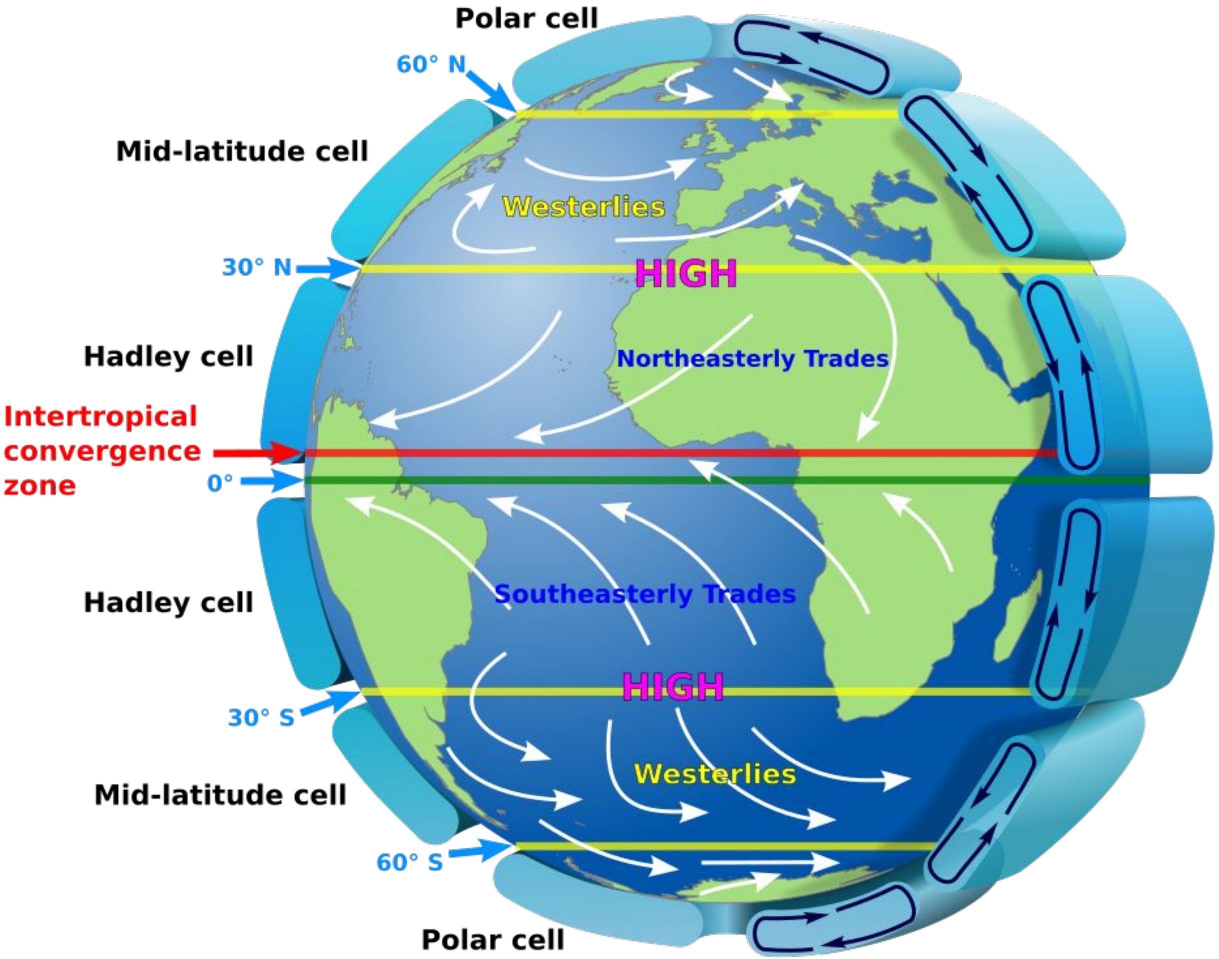
Altitude because it gets cooler as you go up a hill

And the 4th tude:

Attitude If you're in a good mood- it's a sunny day.

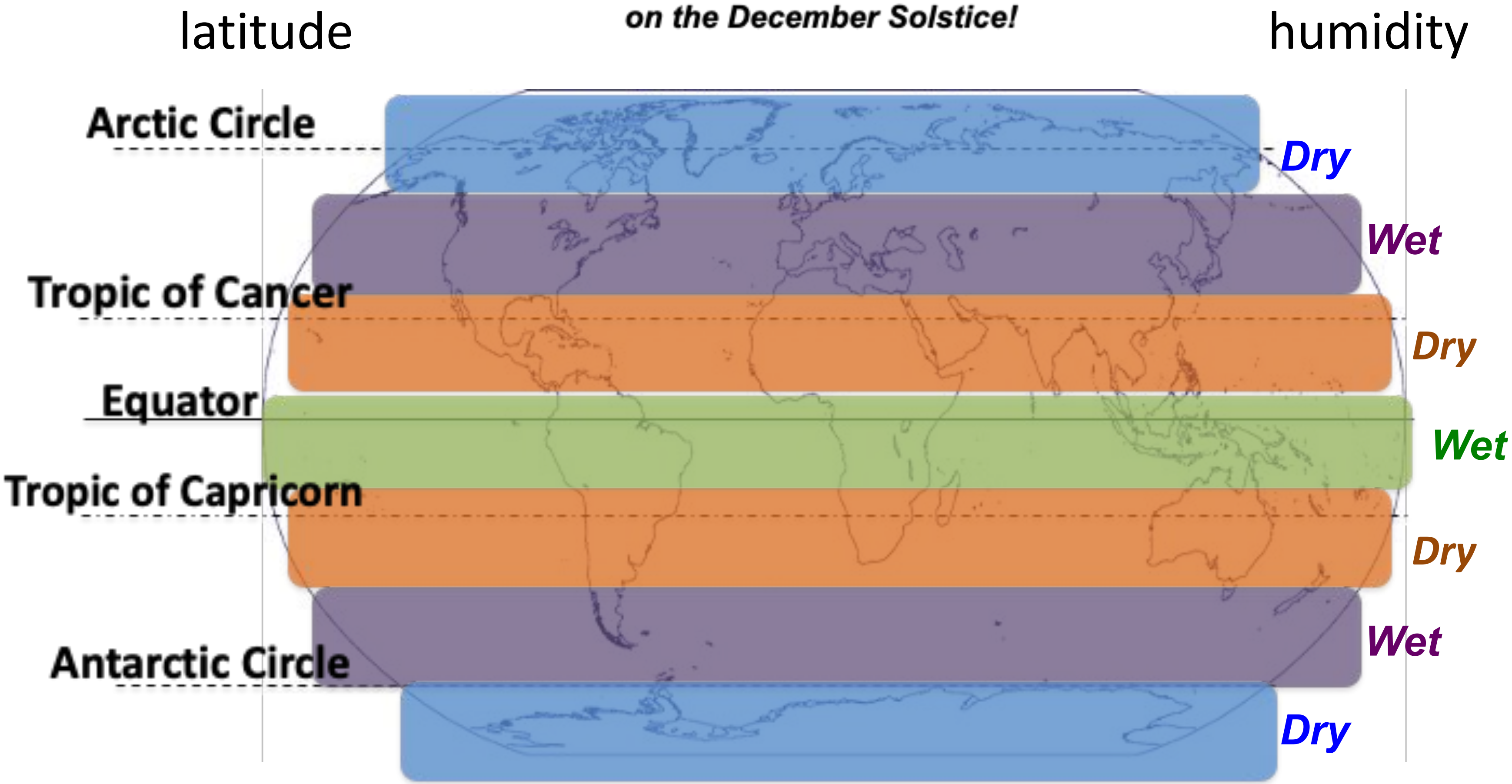
Zones!


latitude
on the equinoxes!
humidity
Arctic Circle
Tropic of Cancer
Equator
Tropic of Capricorn
Antarctic Circle
Dry
Wet
Dry
Wet
Dry
Wet
Dry

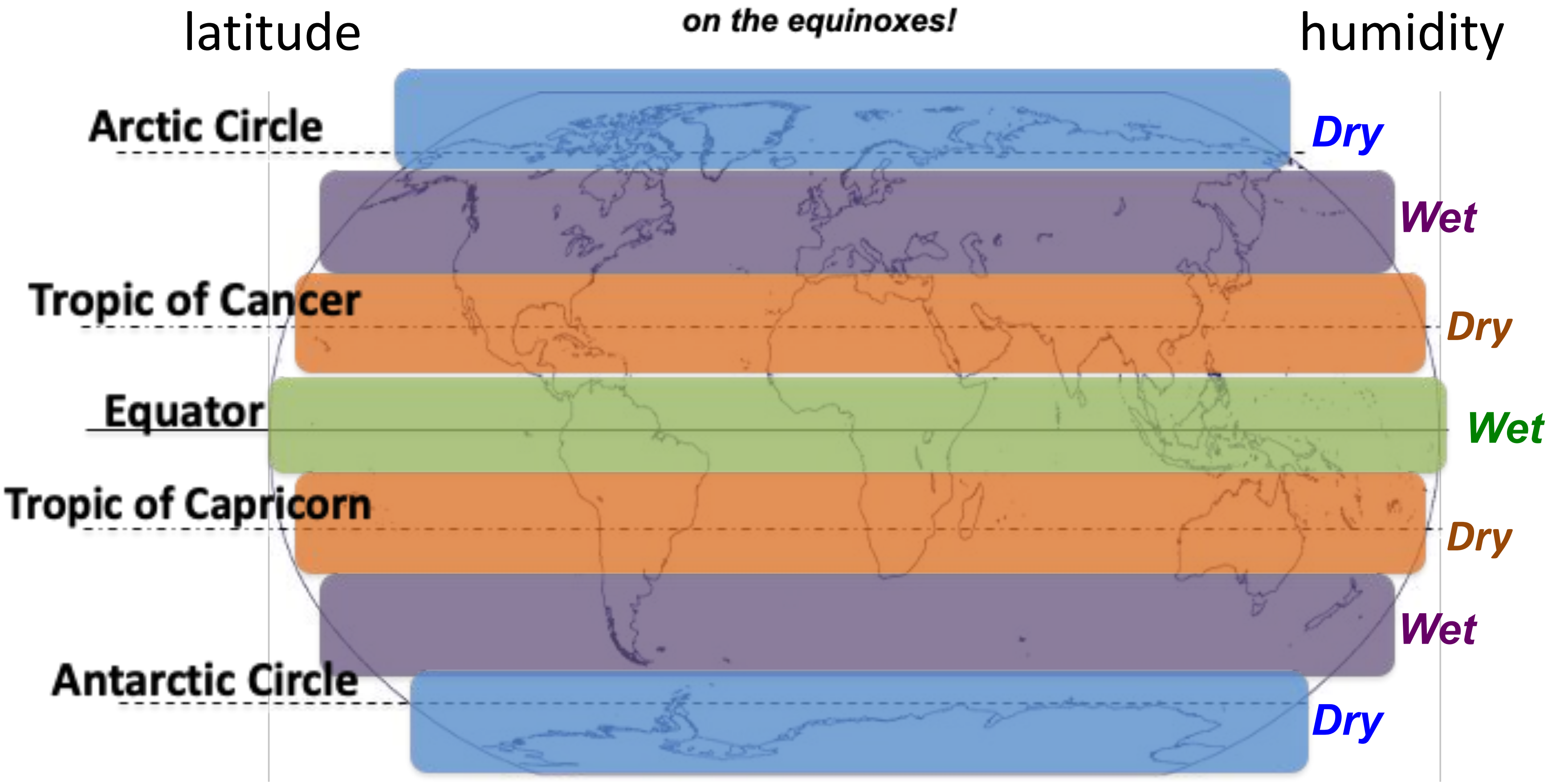

Polar cell
60° N
Mid-latitude cell
Westerlies
30° N
HIGH
Hadley cell
Northeasterly Trades
Intertropical convergence zone
0°
Hadley cell
Southeasterly Trades
HIGH
30° S
Westerlies
Mid-latitude cell
60° S
Polar cell

Zones!

on the December Solstice!



Zones!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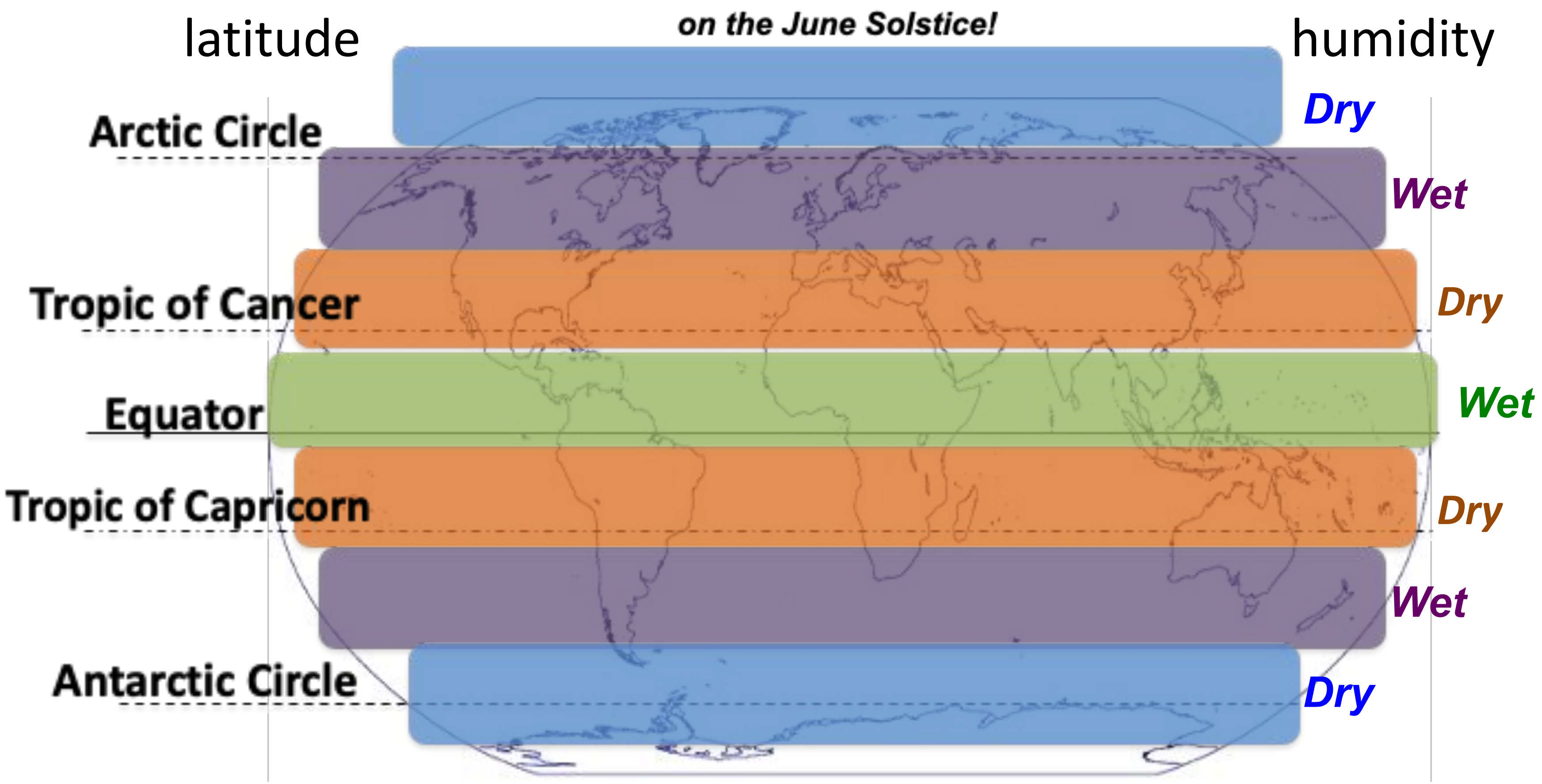
Zones!


latitude
on the June Solstice!
humidity
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Equator
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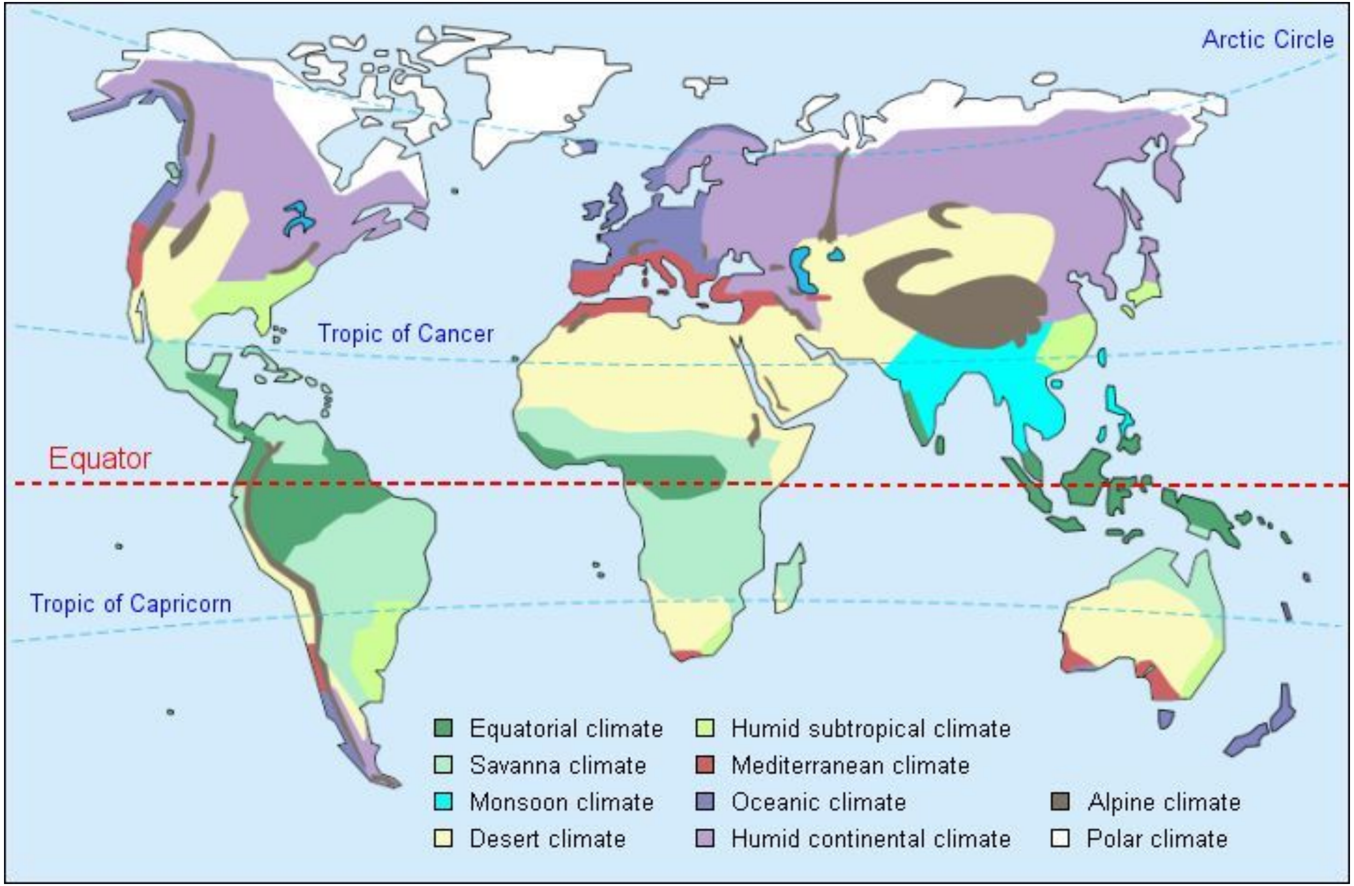
Antarctic Circle
Dry
Wet
Dry
Wet
Dry
Wet
Dry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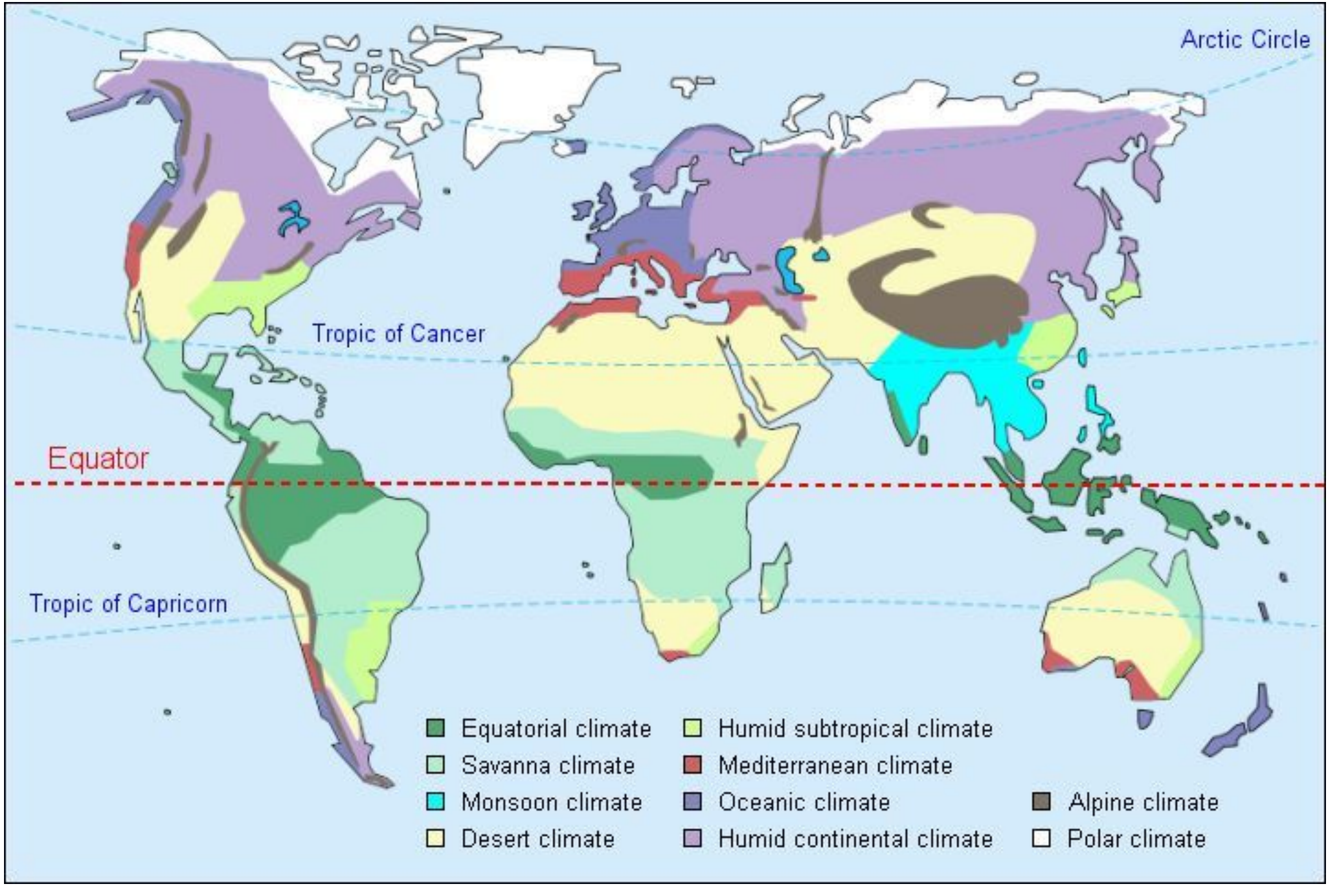
Alpine climate
Polar climate

Rainforest (Moist Tropical)

- ***Temperature:*** ***hot (20º C)***
- ***Humidity:*** ***very wet all year***
- ***Seasons:*** ***rain amount varies***
- ***Latitude:*** ***Equator 0º***
- ***Influence:*** ***n/a***



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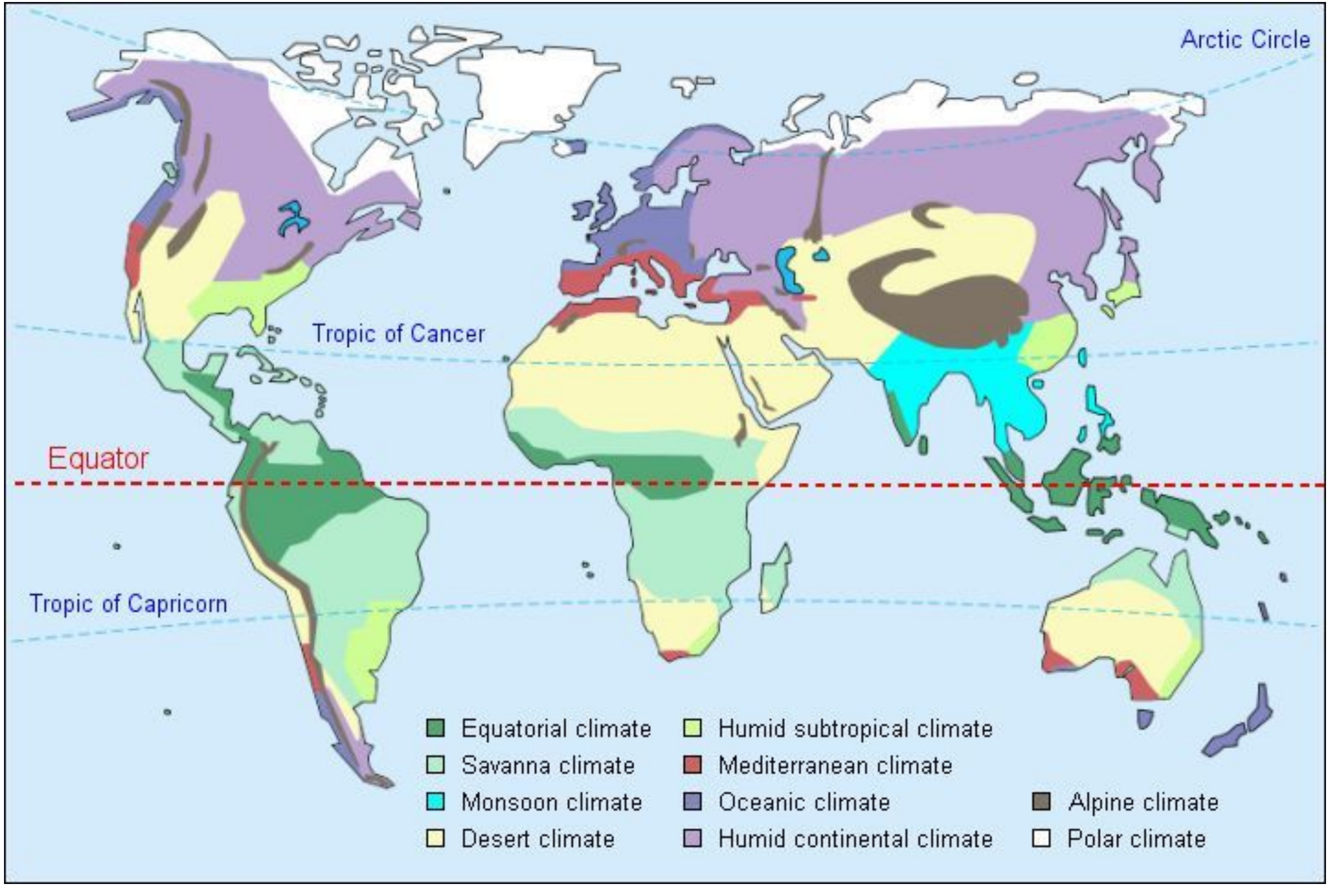
Savannah

- ***Temperature:*** ***hot***
- ***Humidity:*** ***mostly dry***
- ***Seasons:*** ***distinct wet season***
- ***Latitude:*** ***10 – 20º N/S of the Equator***
- ***Influence:*** ***n/a***



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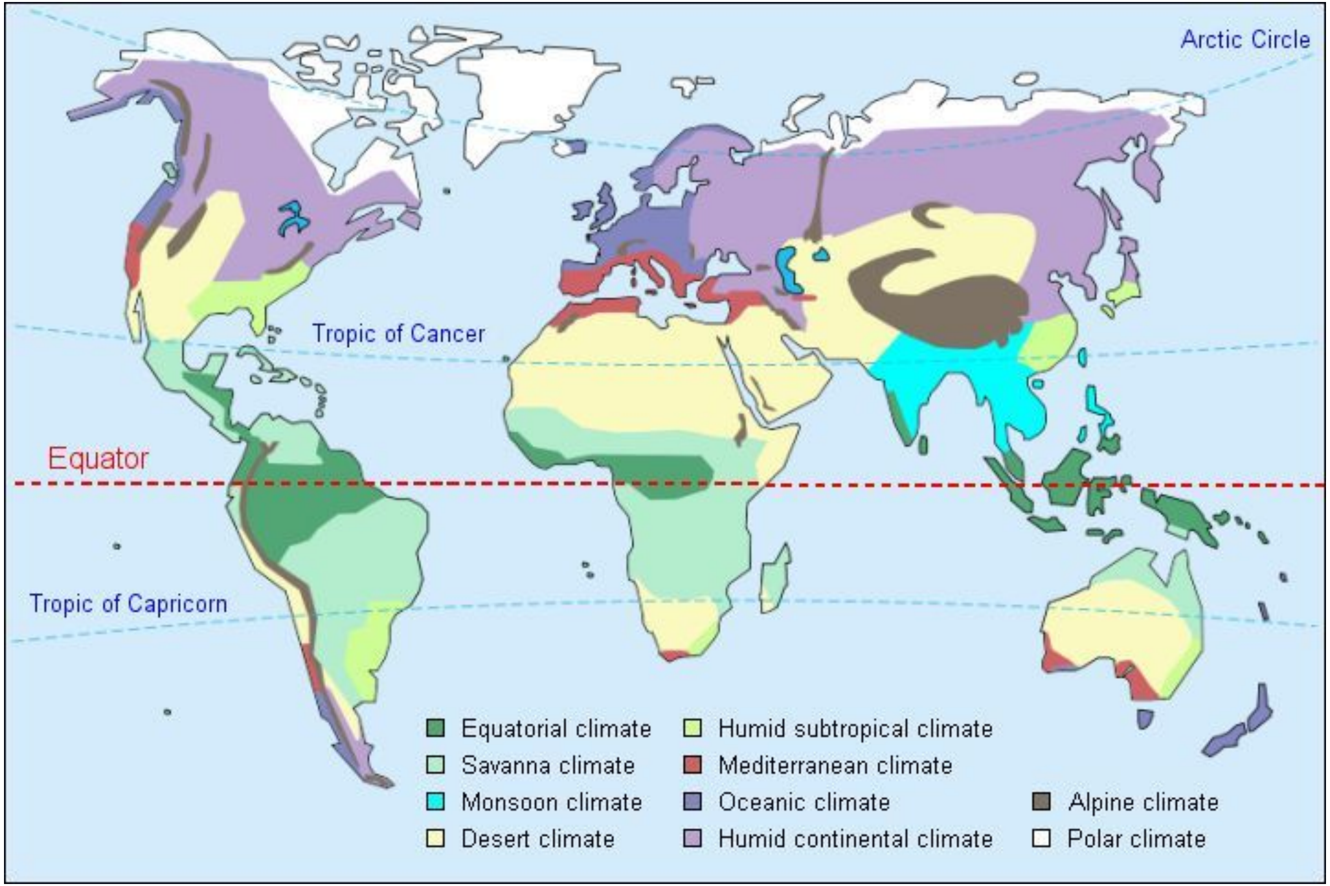
Monsoon

- ***Temperature: hot***
- ***Humidity: mostly wet w/ rain on steroids***
- ***Seasons: distinct dry season***
- ***Latitude: Tropic of Cancer***
- ***Influence: Ocean moisture***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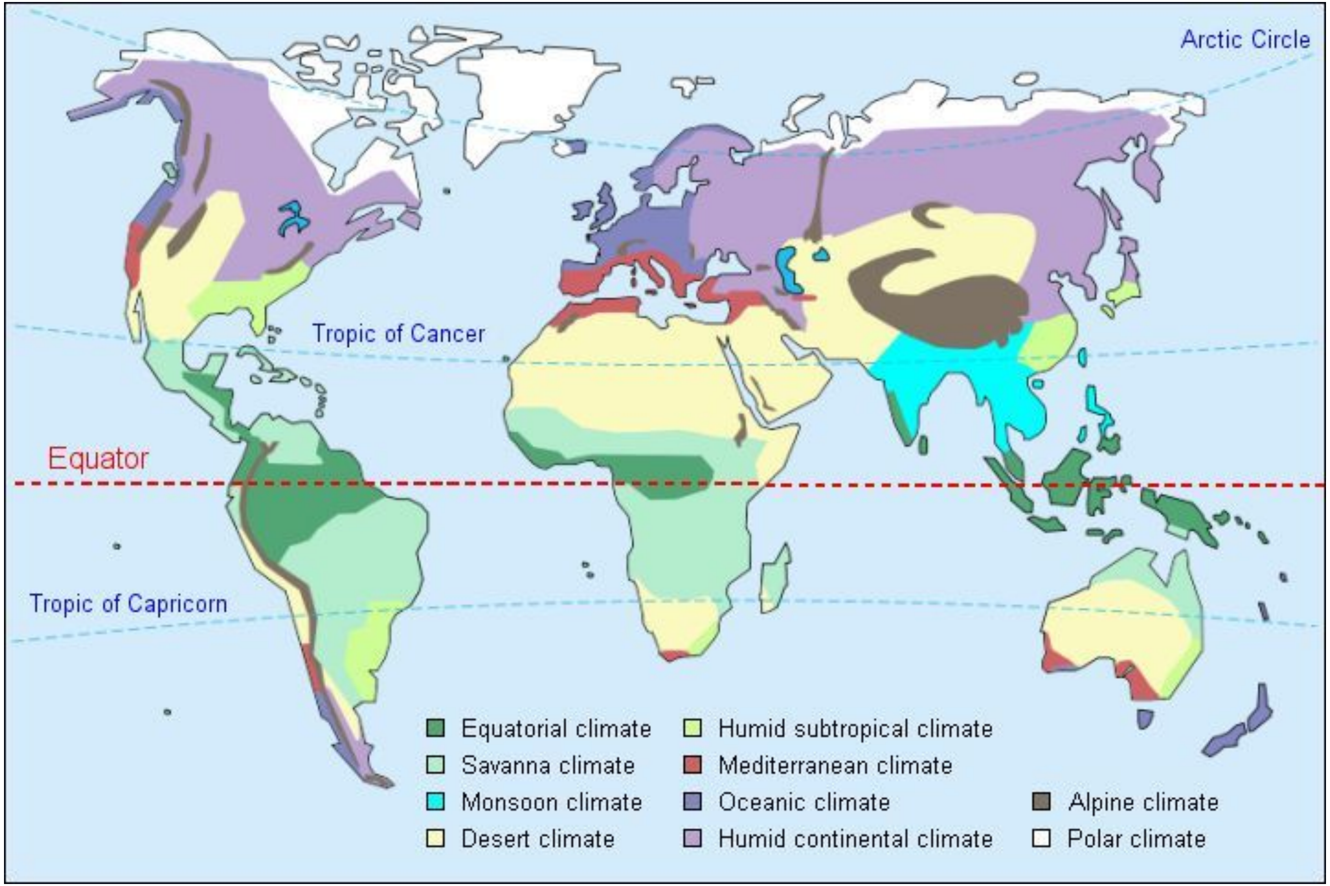
Desert

- ***Temperature:*** *hot and/or cold*
- ***Humidity:*** *dry (arid)*
- ***Seasons:*** *n/a*
- ***Latitude:*** *at the Tropics*
- ***Influence:*** *landmass*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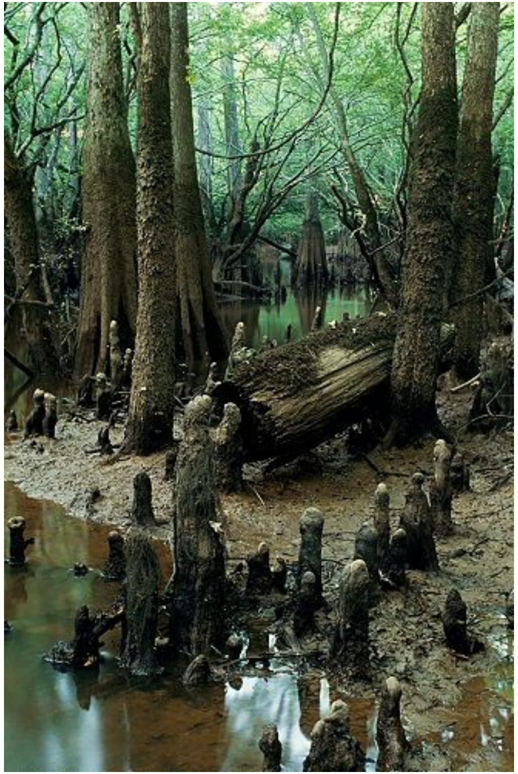
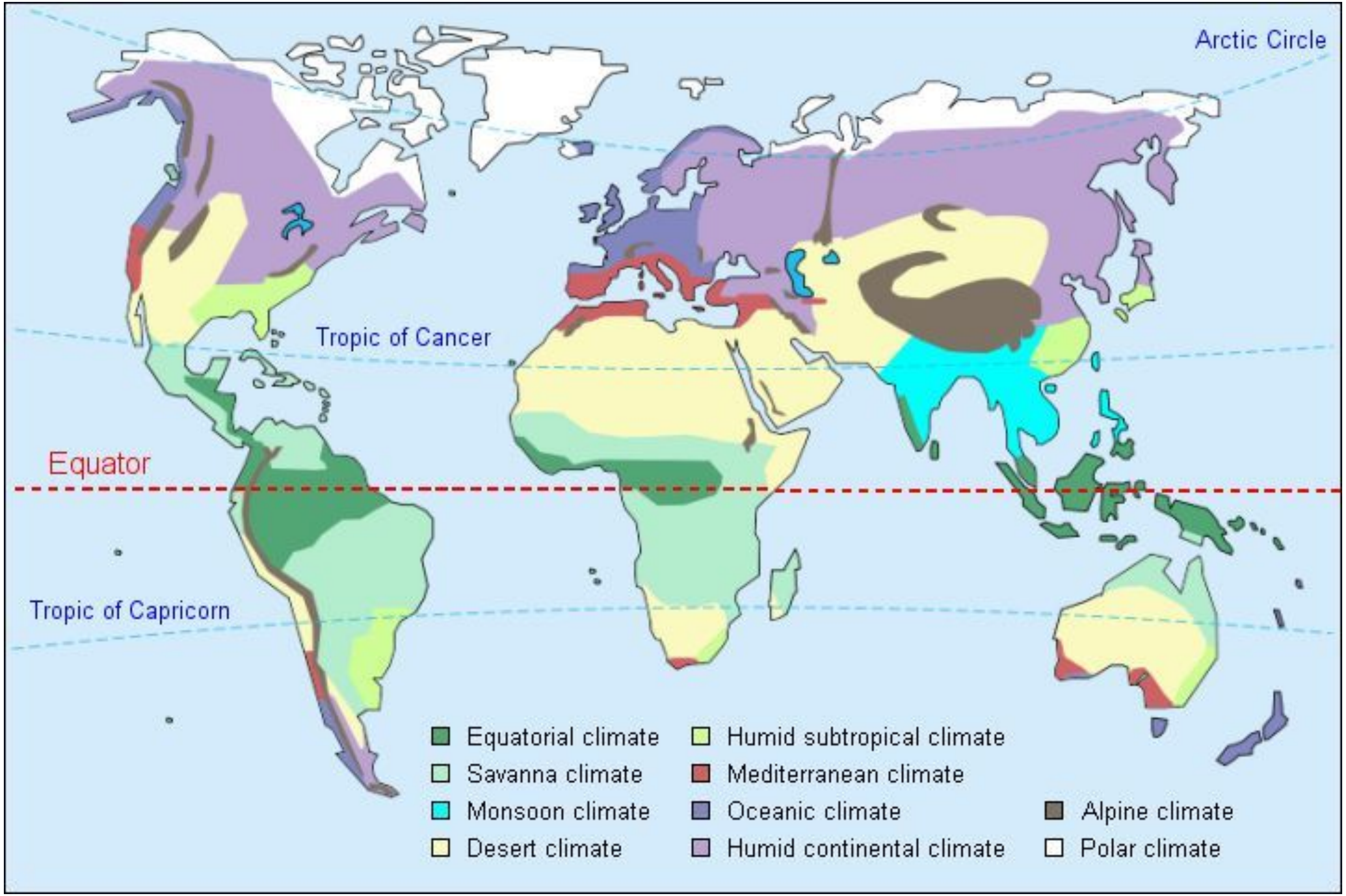
Humid Sub-Tropical

- ***Temperature:*** ***hot summer / mild winter***
- ***Humidity:*** ***wet all year***
- ***Seasons:*** ***mild winter / hot wet summer***
- ***Latitude:*** ***Sub-Tropics 30 – 45º***
- ***Influence:*** ***East on a landmass***

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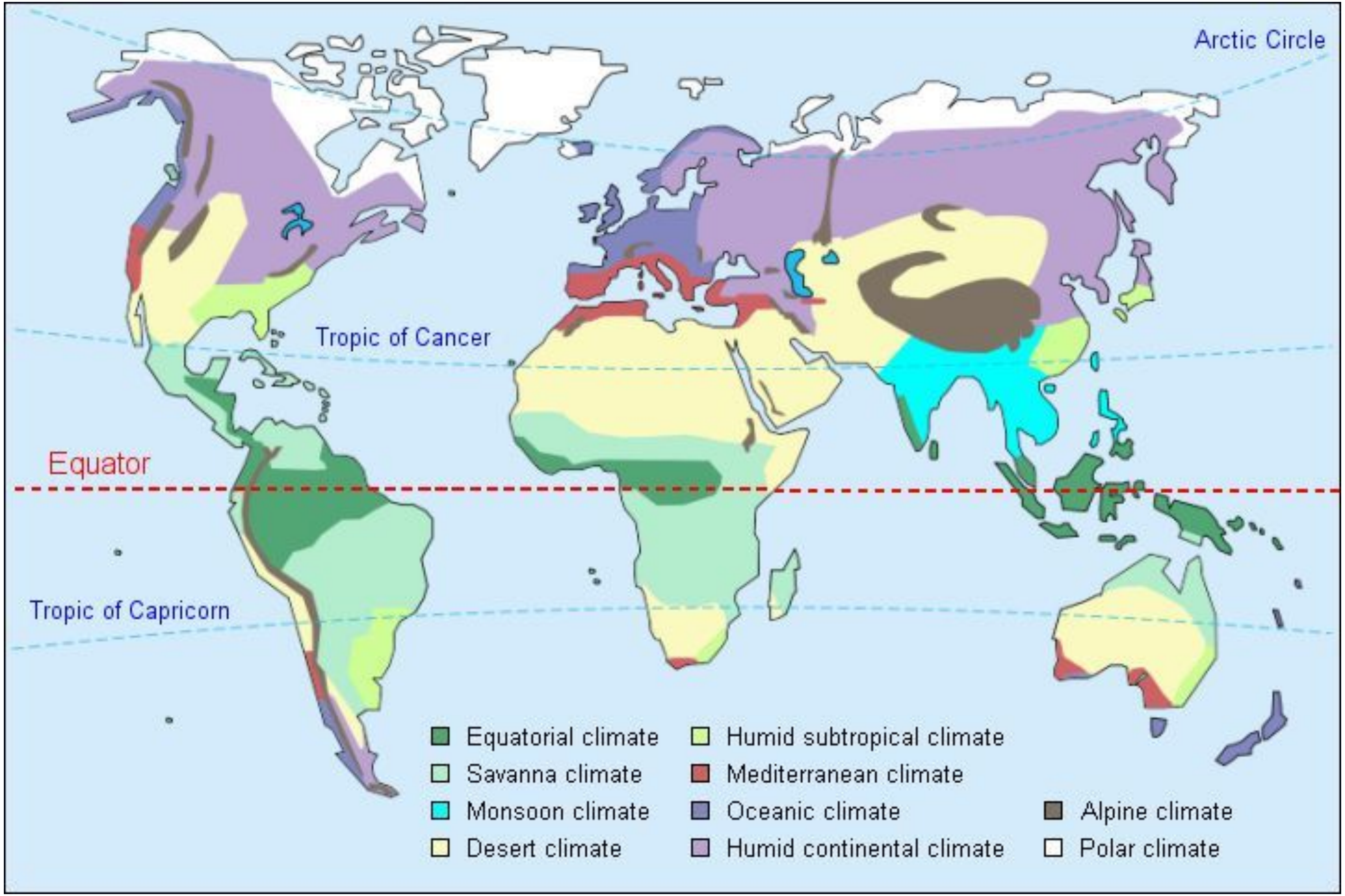
Mediterranean

- ***Temperature:*** ***warm and cool***
- ***Humidity:*** ***dry***
- ***Seasons:*** ***mild wet winter / dry summer***
- ***Latitude:*** ***Sub-Tropics 30 – 45º***
- ***Influence:*** ***ocean / west coast of a landmass***



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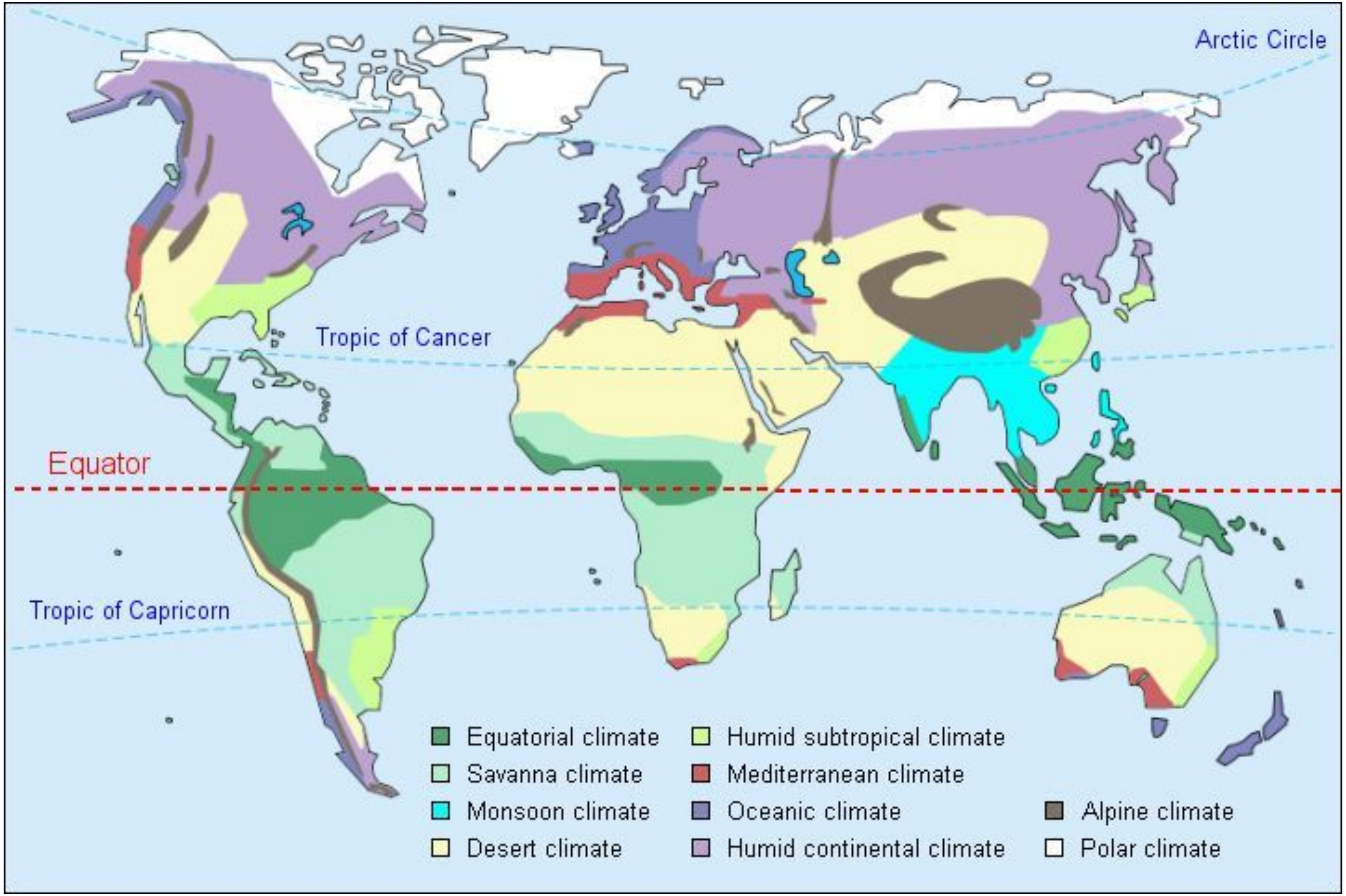
Oceanic

- ***Temperature:*** ***mild/cold***
- ***Humidity:*** ***wet***
- ***Seasons:*** ***mild/cool variation 10 to 15º spread***
- ***Latitude:*** ***middle 45 – 65º***
- ***Influence:*** ***ocean/ west coast***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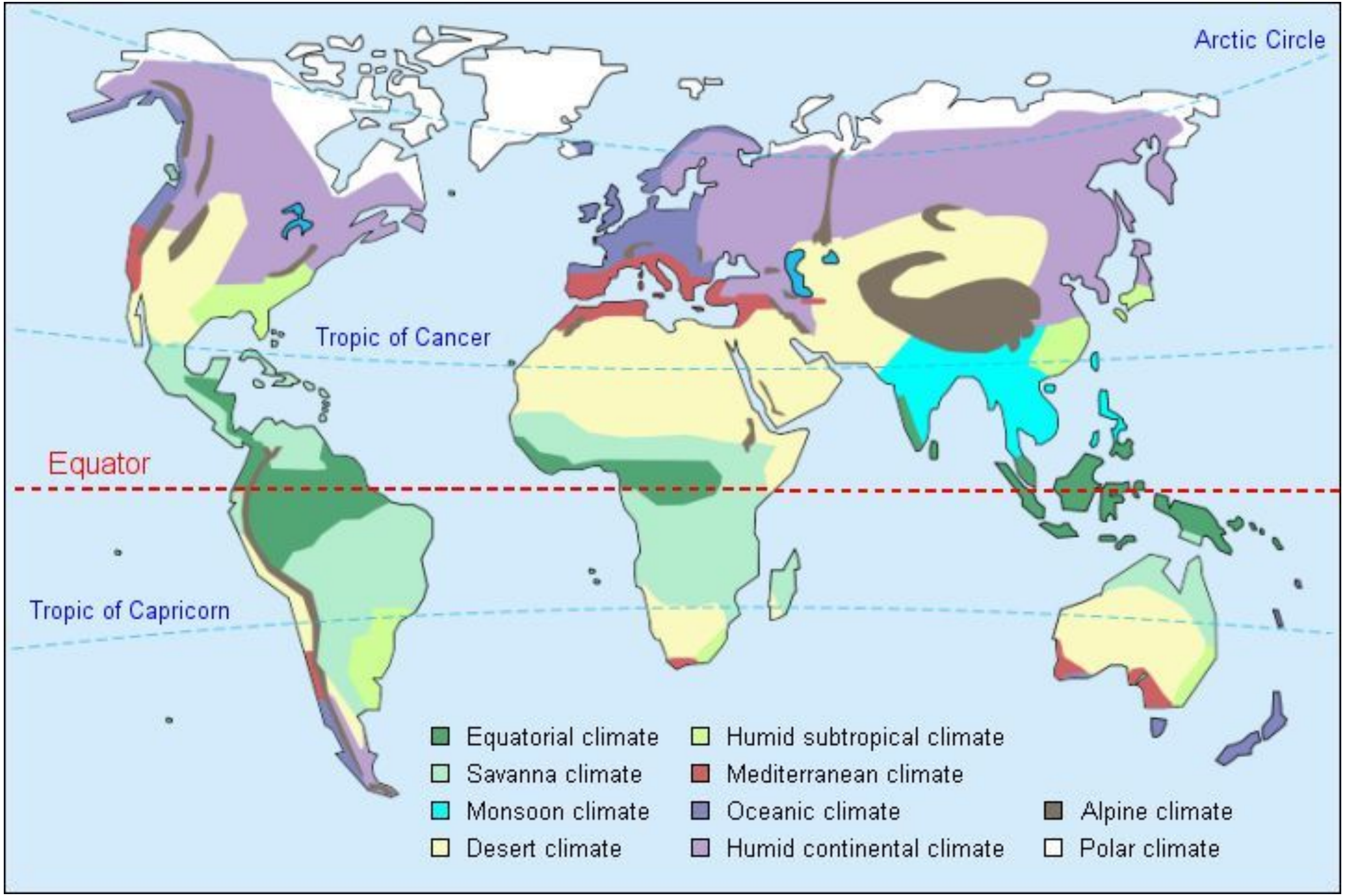
Humid Continental

- ***Temperature:*** ***extremes 20º+***
- ***Humidity:*** ***wet year-round***
- ***Seasons:*** ***extremes***
- ***Latitude:*** ***middle 45 – 65º***
- ***Influence:*** ***landmass***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Alpine

- ***Temperature:*** ***extremes***
- ***Humidity:*** ***wet and dry***
- ***Seasons:*** ***extremes***
- ***Latitude:*** ***all***
- ***Influence:*** ***altitude***

Polar

- ***Temperature:*** ***cold***
- ***Humidity:*** ***dry***
- ***Seasons:*** ***light/dark***
- ***Latitude:*** ***poles 70° +***
- ***Influence:*** ***n/a***

Climate:	Temperature	Humidity	Seasons	Latitude	Influence
Rainforest	Hot	wet	None - sometimes less wet	equator	n/a
Savanna	hot	Dry with a distinct wet season	3-4 months of wet	10º to 15º N/S	n/a
Monsoon	hot	Wet with a distinct dry season	Wettest in the hottest season	At the Tropic of Cancer	Ocean moisture
Desert	Hot or cold	dry	Only temperature varies	At the Tropics and the poles	In the shadow of a mountain range
Humid Sub-Tropical	Hot and cool	wet	Mild winters, hot, wet summer	Sub-Tropics 25º to 45º N/S	East coast of a landmass

Climate:	Temperature	Humidity	Seasons	Latitude	Influence
Mediterranean	Hot and cool	Mostly dry with a wet season	Mild wet winter w/ dry summer	Sub-Tropics 25º to 45º N/S	West coast of a landmass
Oceanic	Mild range	wet	Cold, rainy winters	Mid latitudes 45º to 65º N/S	West coast of a landmass
Humid Continental	Extremes hot and cold	wet	Extremes all four seasons	Mid latitudes 45º to 65º N/S	East and center of a landmass
Alpine	Colder than would be at each latitude	Wet to dry	Extremes	any	elevation
Polar	cold	dry	Light or dark	At the poles	n/a

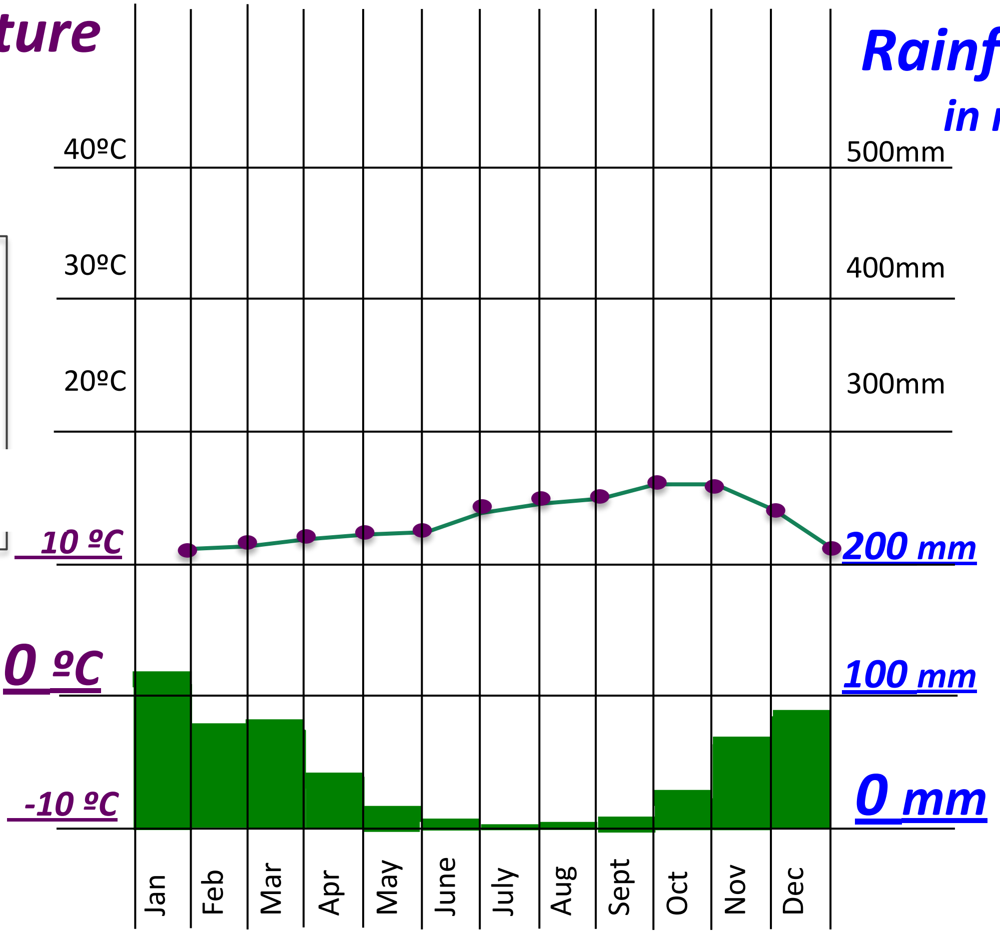
Climate Graph

Temp range:
11º to 18º C

Rainfall range
5mm to 110mm

San Francisco

Lat: 37N
Long: 122W

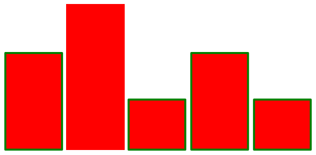


Directions:

- Draw a box around or highlight the data rows for your city.

- Write the name of your city at the top of the graph.
- Write the latitude and longitude at the top of the graph.
- Make a dot to dot graph to represent the temperature using the scale on the left side of your graph (ºC)

- Make a bar graph to represent the rainfall using the scale on the right side of your graph (mm) in the **COLOR** your slip indicates.



At top:

City Name

Lat: _______

Long: _______

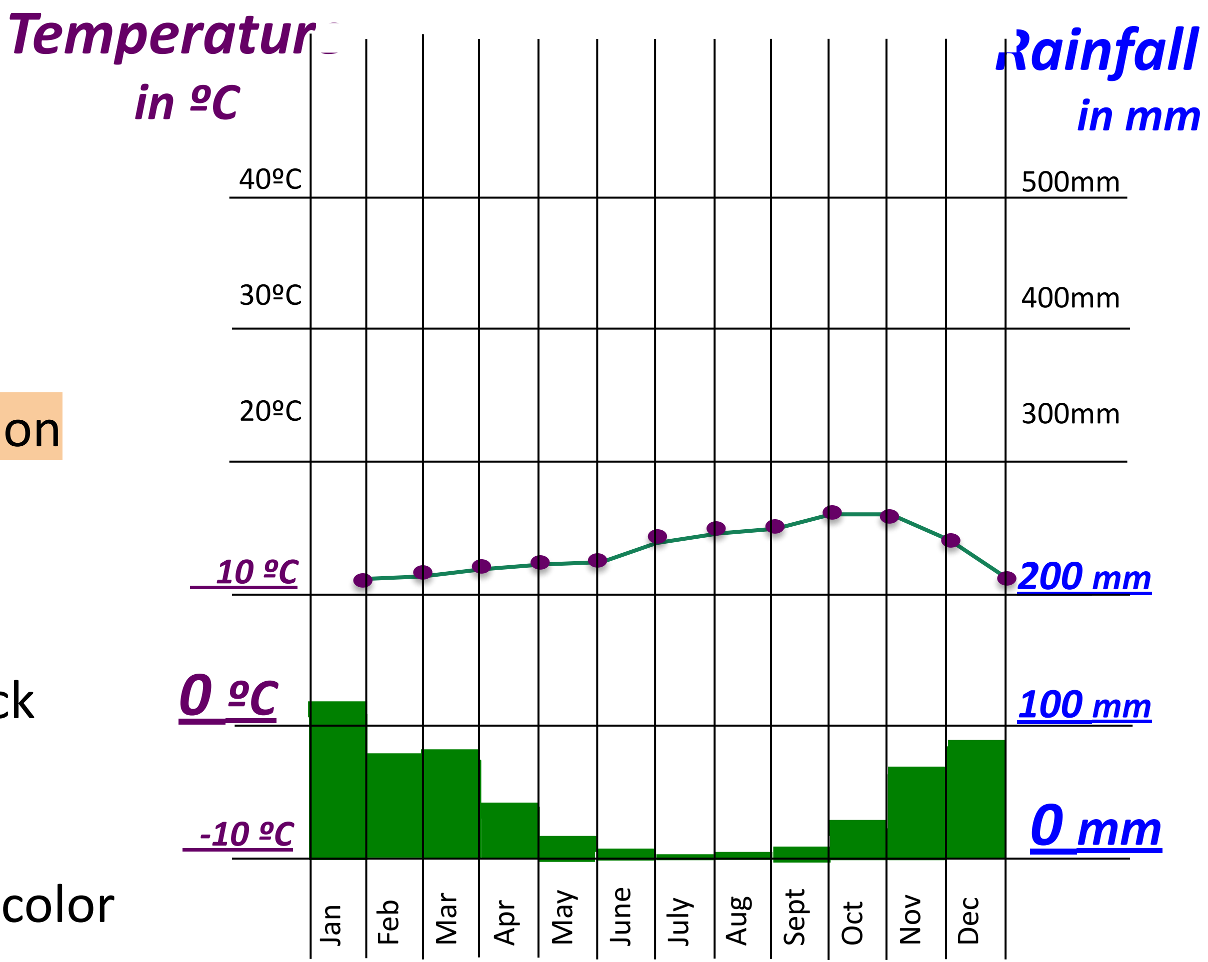
Highlight both rows on data sheet

ºC - Temperature:

dot to dot in black

mm - Rainfall:

bar graph in city color



Early finishers: use your chromebooks to do some more research on your city.

- What country is it in? What language do they speak there?
- What is the climate of the country as a whole? Is the city's climate a good representation of the country's climate?
- What are the historic high and low temperatures?
- What sort of activities can you do in your city? Are there any cultural events (festivals, holidays) that happen during the months where it would be nice to travel there?

Discussion Questions

1. What is the name of your city? What country is it in?
2. What climate does your city have? Describe the temperature and precipitation across the year – describe the climate season by season.
3. Describe how the 'tudes (or any other factors) influence the climate there.
4. Find another city graphed by one of your classmates that has a similar climate. How similar is it to your city's climate? Where is it on the globe compared to your city?

YOU MAY USE YOUR CHROMEBOOK TO CONVERT CELSIUS AND MILLIMETERS INTO UNITS THAT YOU UNDERSTAND